Guichon Creek Raven HBV-EC Model: Baseline Summary

m.ah a temEEwuh Solar Project, land-cover effects on Guichon Creek flows (Swiftwater SWC-1-43-01-P-001). Baseline model as of git commit e152a45 (2026-11-09). Report differences between runs, not absolute flows.

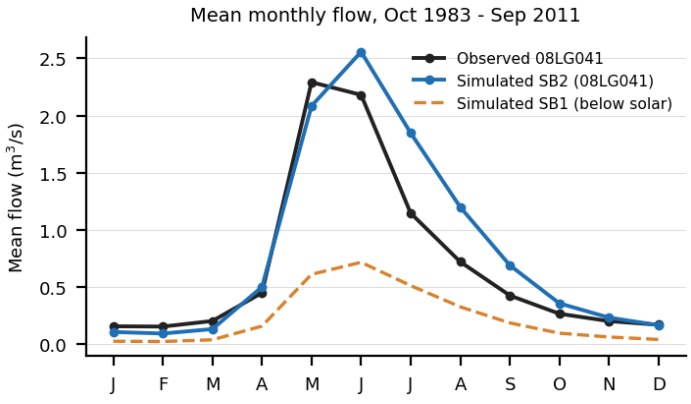
Model set-up

Item	Setting
Model	Raven v4.1, HBV-EC emulation (Manual App. F.2), daily step
Run period	1982-10-01 to 2011-09-30. Spin-up WY1983; calibration Oct 1983 - Sep 1999; validation Oct 1999 - Sep 2011
Forcing	ECCC Highland Valley Lornex (1123469, 1268 m), gaps from Kamloops A. P = 382 mm/yr basin mean
PET	PET_FROMMONTHLY: Hargreaves monthly normals at gauge (~713 mm/yr)
Orographic	OROCORR_HBV: temperature lapse 6.5 C/km; precipitation fixed +8%/km
Routing	Triangular unit hydrograph (HBV MAXBAS) in each subbasin; ROUTE_NONE between subbasins
Mamit Lake	Lake-like reservoir at SB2 outlet: area 2.03 km^2, weir C 0.6, crest 10 m. No lake HRU (no lake evaporation)
Observed	WSC 08LG041 at Mamit Lake outlet. **Regulated** (Mamit Lake dam rebuilt 1985, irrigation, Highland Valley mine)
Scenario	:LandUseChange / :VegetationChange on SOLAR_SITE (HRUs 1, 6), dated 1900-01-01 (applied at t = 0)

Subbasins and HRUs (11)

ID	Basin	Band	Area km^2	Elev m	Land cover (pre-project)
1	SB1	Solar	3.24	1215	SITE_PRE: 50% forest
2-5	SB1	1000-1600+	181.58	1153-1719	OPEN: 0% forest
6	SB2	Solar	0.75	1232	SITE_PRE: 50% forest
7-11	SB2	<1000-1600+	619.41	974-1670	MIX20: 20% forest

SB1 = Guichon_BelowSolar (184.8 km^2, outlet below solar site: the answer). SB2 = Guichon_08LG041 (620.2 km^2 incremental, calibration). Bands: <1000, 1000-1200, 1200-1400, 1400-1600, 1600+ m. Solar run: HRUs 1, 6 change to SOLAR (0% forest, IMPERM 0); solar-site parameters are not calibrated.



Parameters

Parameter (HBV name)	Value	Source
RAINSNOW_TEMP (TT)	0.5 C	Cal. step 1
RAINSNOW_DELTA (TTI)	2.0 C	Default
ADIABATIC_LAPSE (TCALT)	6.5 C/km	Cal. step 1
MELT_FACTOR (CFMAX)	2.5 mm/d/K	Cal. step 1 (was 4.0)
MIN_MELT_FACTOR	1.5 mm/d/K	Cal. step 1 (was 2.2)
HBV_MELT_FOR_CORR (MRF)	0.5	Default
REFREEZE_FACTOR (CFR)	2.0 mm/d/K	Default
SNOW_SWI (CWH)	0.05	Default
TOPSOIL thickness x POROSITY (FC)	2.0 m x 0.2 = 400 mm	Cal. step 3 (was 200 mm)
FIELD_CAPACITY (LP)	0.5	Cal. step 3 (was 0.7)
HBV_BETA (BETA)	2.0	Tested step 3, unchanged
MAX_CAP_RISE_RATE (CFLUX)	1.0 mm/d	Default
MAX_PERC_RATE (PERC)	5.0 mm/d	Tested step 2, unchanged
FAST_RES BASEFLOW_COEFF (KF)	0.05 1/d	Cal. step 2 (was 0.2)
FAST_RES BASEFLOW_N (1+ALPHA)	1.5	Default
SLOW_RES BASEFLOW_COEFF (KS)	0.02 1/d	Cal. step 2 (was 0.05)
TIME_CONC / TIME_TO_PEAK (MAXBAS)	SB1 2.0/1.0 d; SB2 3.0/1.5 d	Tested step 2, unchanged
:RainCorrection / :SnowCorrection	1.0 / 1.0	Not calibrated
Reservoir CrestWidth / WeirCoeff.	10 m / 0.6	Not calibrated

Calibration (manual, 3 steps, grid searches)

1. Snow timing (72 runs): melt factors, TT, lapse. 2. Response shape (144 runs): KF, KS, PERC, TIME_CONC. 3. Volume (100 runs): LP, BETA, TOPSOIL. Steps 1-2 scored on volume-independent fit because the gauge is regulated.

Metric (08LG041)	Cal 1983-99	Val 1999-2011	All
NSE	-0.04	0.57	0.19
KGE	0.32	0.60	0.61
PCT_BIAS	+45%	-18%	+13%
NSE, sim rescaled to obs volume	0.45	0.56	-
Before calibration: NSE / PCT_BIAS (all)			-6.54 / +158%

Limitations

- **Regulated gauge.** The dam stores the freshet (fills by late June) and releases water for irrigation. The model simulates natural flow, so it over-predicts June-September (storage and irrigation). Bias is a deliberate compromise (cal +45%, val -18%), not zero.
- **Non-stationary record.** Every parameter set shows a 60-75 point bias gap between periods (dam rebuilt 1985).
- **Water-limited.** ET ~351 of 382 mm/yr, so runoff is a small residual. The solar delta is sensitive to ET parameters: bracket it with LP 0.4 and 0.6.
- **Not yet included:** lake evaporation, :LateralEquilibrate (shared HBV zones), channel routing, real slope/aspect. SB1 has no gauge for direct validation.